

Mechanical engineer with hands-on experience managing the full product development cycle for EV battery pack systems — from design through prototype evaluation, supplier coordination, and mass-production launch. Native English speaker (Indian, educated in English) and JLPT N1-certified in Japanese, enabling direct bilingual communication with clients, suppliers, and overseas partners. On an outsourced project, served as the primary communication point between the client, multiple client-side departments, and prototype/mass-production makers — coordinating requirements across all parties while simultaneously executing the engineering design work. Proven ability to iterate designs based on evaluation feedback to meet cost, weight, and manufacturability targets.

EDUCATION

Kyoto University - Bachelor of Engineering in Mechanical Engineering, Kyoto, Japan Apr 2019 — Mar 2023

- TOPIC : Aluminum corrosion behavior in ionic liquid electrolytes for lithium secondary batteries.
Conducted undergraduate thesis research on enhancing the safety and performance of lithium-ion batteries using flame-retardant ionic liquid electrolytes. Focused on the corrosion behavior of aluminum current collectors in Li[FSA]-based systems, analyzing the effects of temperature and lithium salt concentration through electrochemical and surface characterization techniques.

Osaka University - Japanese Language Extensive Course, Osaka, Japan Apr 2018 — Mar 2019

CERTIFICATES

JLPT N1, The Japan Foundation and Japan Educational Exchanges and Services Jul 2021
TOEIC 920, ETS Aug 2023

TECHNICAL EXPERIENCE

Mechanical Engineer / Chassis, Aluminium extrusion Parts Design-BEV Development April 2026 — Present
Daihatsu Motor Co., Ltd. (via Persol Cross Technology) Osaka, Japan

- Design chassis components for a new BEV (battery electric vehicle) development project.
- Design energy-absorbing (EA) material structures to protect the battery pack from shock and collision loads.
- Perform clearance calculations to ensure structural and packaging requirements are met.
- Conduct 3D part design in collaboration with suppliers while concurrently working on chassis-level integration.
- Apply CATIA V5 3D/2D modeling, CAE structural analysis, and cross-functional communication skills across both chassis and battery-protection design work.

Mechanical Engineer / Sheet Metal Parts Design-HEV Development May 2024 — March 2026
Daihatsu Motor Co., Ltd. (via Persol Cross Technology) Osaka, Japan

- Designed and developed battery pack systems for next-generation electric vehicles.
- Created battery pack mounting structures, considering component layout, structural strength, and mounting clearances. Conducted CAE analysis to evaluate the strength and durability of designed components.
- Designed surrounding sheet metal and plastic parts using CATIA V5, optimizing shape, structure, and spatial integration.
- Collaborated with suppliers to manufacture prototype sheet metal parts and coordinated with evaluation teams to assess performance. Iterated designs based on evaluation results to meet functional and structural requirements.
- Worked with mass production suppliers to assess part formability, incorporating changes to reduce cost and weight. Finalized part design based on functionality and manufacturability, and validated mass production prototypes.

Mechanical Engineer / HVAC Piping Design Dec 2023 — Apr 2024
Daihatsu Motor Co., Ltd. (via Persol Cross Technology) Osaka, Japan

- Designed and optimized piping systems for vehicle air conditioning, focusing on efficient layout and structural integration.
- Calculated variations due to hose length, bending radius, routing complexity, and engine dynamics to ensure reliable performance and established piping routes while maintaining required clearances in a compact engine bay environment.

Mechanical Engineer / Freshly Graguated Traning Apr 2023 — Nov 2023
Persol Cross Technology Nagoya, Japan

- Trained in CATIA V5 solid/surface modeling and core mechanical design fundamentals.

SKILLS

Tools and Technologies	CATIA V5 3D & 2D, Microsoft PowerPoint, Excel, VS Code
Analysis	Primary & secondary strength analysis, Fatigue Analysis, Meshing, Testing & Validation, Failure Mode and Effect Analysis (FMEA)
Programming Languages	Python, JavaScript, HTML, CSS
Courses Taken	Sheet metal bending and forming, Strength of Materials

ACHIEVEMENTS

MEXT Scholarship recipient One of only five students selected nationwide for overseas study in Japan (2018–23).